FACULTY OF INFORMATION TECHNOLOGY
VNU UNIVERSITY OF ENGINEERING AND TECHNOLOGY - VNU

MULTI-OBJECT MULTI-CAMERA TRACKING SYSTEM

STUDENT RESEARCH

Team members: Le Duy Son - 19020418
Tran Van Trong Thanh - 19021367
Luong Son Ba - 20020001
Trinh Ngoc Huynh-20020054
Vu Huy Hoang -20020414

Supervisors: Dr. Tran Quoc Long

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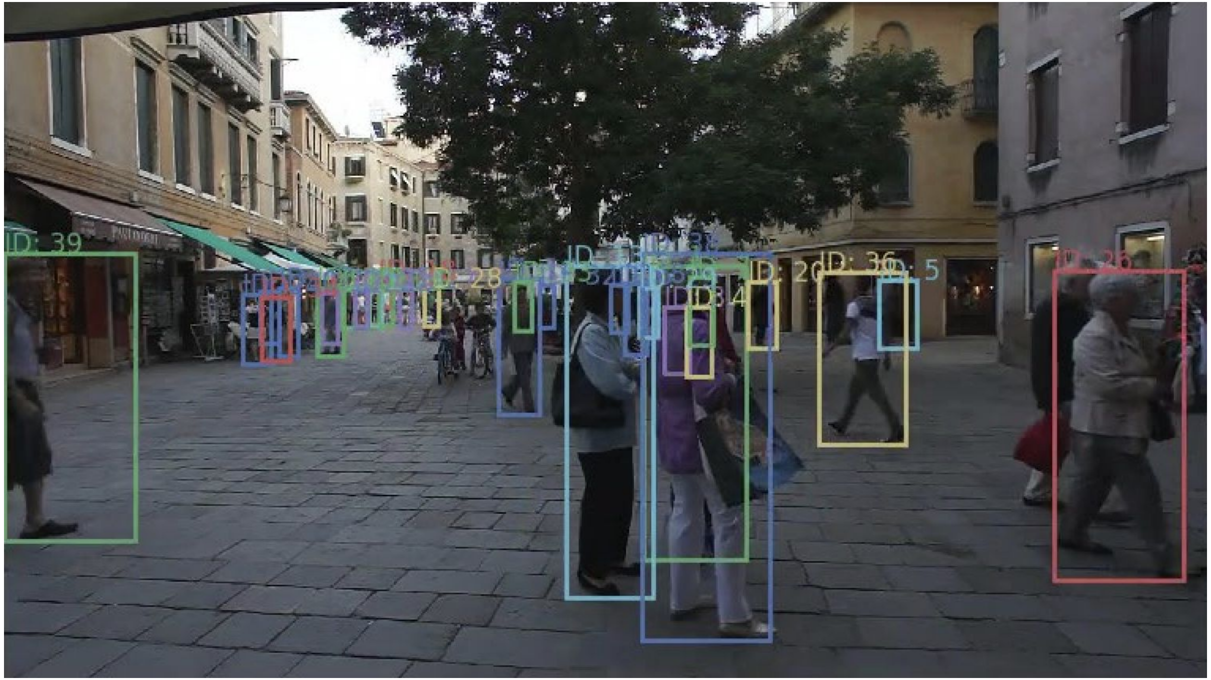
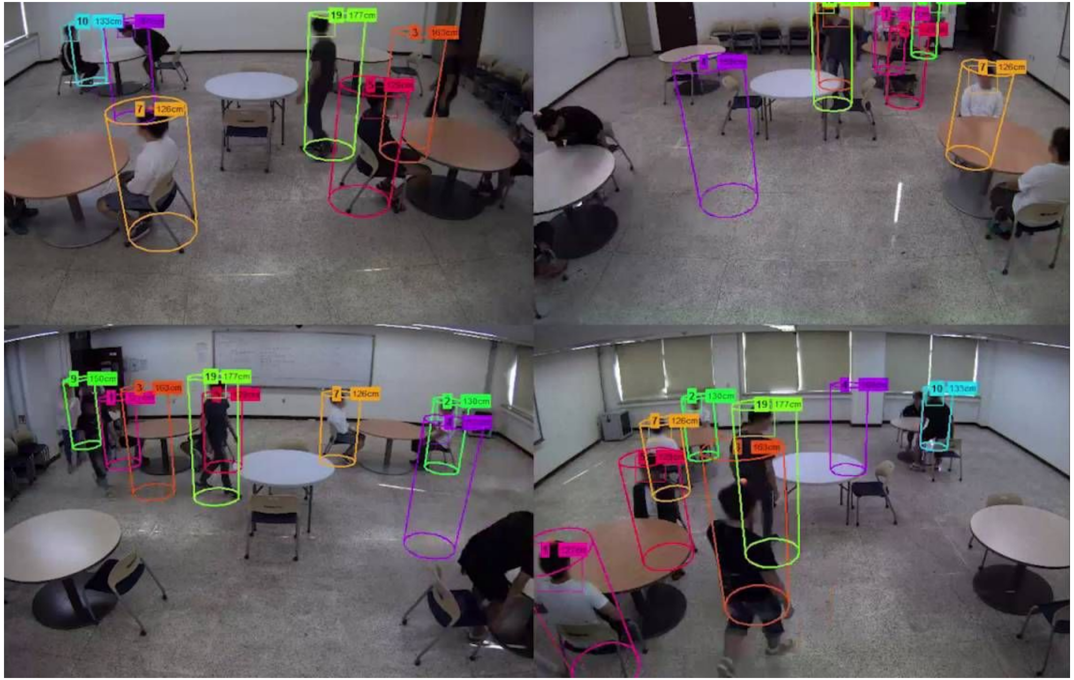
INTRODUCTION

1.1 | A general view of Tracking



Example of MOT

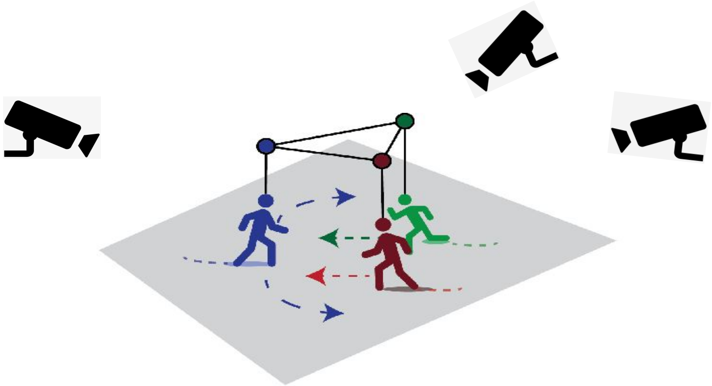
1.2 | Multiple targets - Multiple cameras Tracking



Example of MTMC

1.3 | Contributions

- Combine re-identification and **Pose Estimation** improve feature representation.
- Using **QDTrack's training strategy** to improve re-identification.
- Adapt solution **Observed Centric** to solve the problem of Kalman Filter (KF).
- **Online MTMC pipeline** for human surveillance in the workplace.
- Achieve **real-time performance** for MTMC problems.



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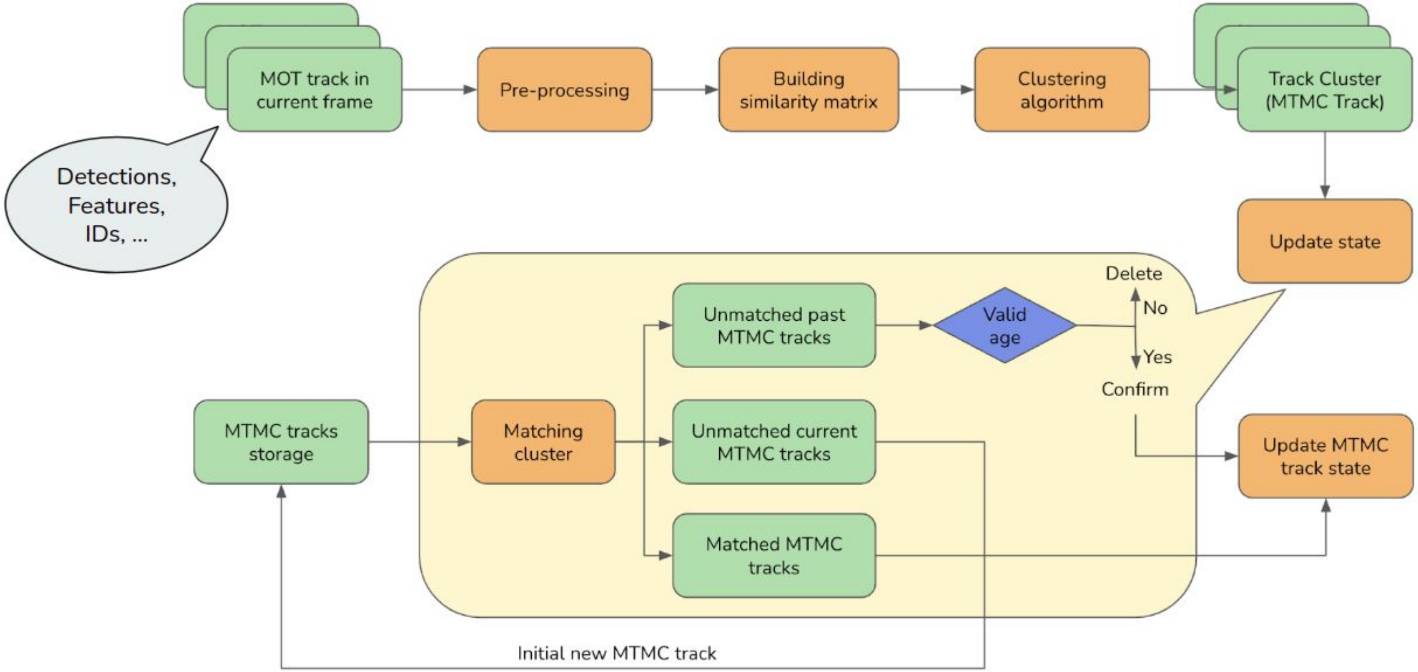
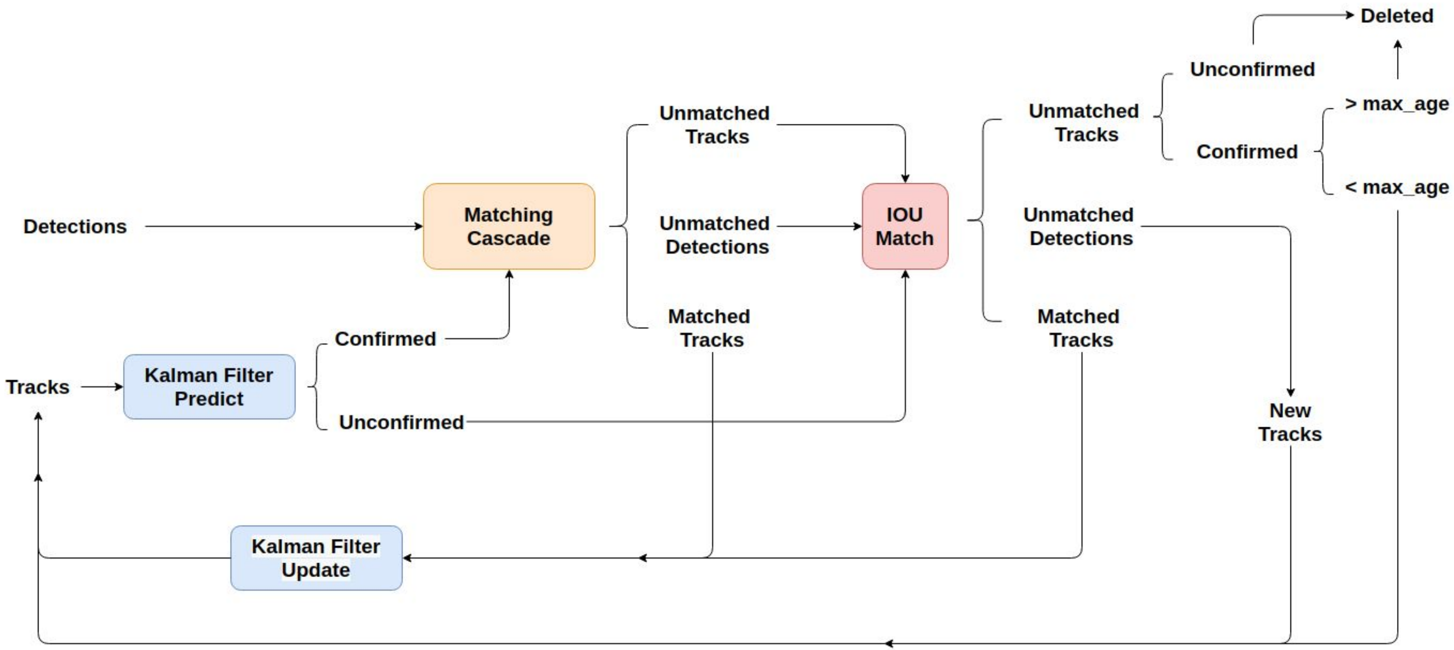
BACKGROUND



MTMC system pipeline



DeepSORT pipeline

2.2 | Drawback of MTMC

- Clustering solely based on original distance ⇒ **Inaccurate** data association.
- Too rely on ReID features ⇒ **Struggle** in case of occlusion, similar appearance, etc.
- Many current systems still **under-perform** in real-time tracking.

2.2 | Drawback of MTMC

MTMC

- **Inaccurate** ranking distance
- **Struggle** when similar object appearance
- **Under-perform** in real-time tracking

2.3 | Drawback of DeepSORT

- Requires **a high frame rate** for the video input.
- Kalman Filter (KF) is **sensitive to motion noise**.
- When objects are occluded or have non-linear motion, there is no detection to update KF, resulting in **accumulated errors**.
- ReID model in DeepSORT **does not have many samples** to learn how to extract appearance features of object effectively.

2.3 | Drawback of DeepSORT

DeepSORT

- **High frame rate** for the video input
- Kalman Filter (KF) is **sensitive to motion noise**
- **Accumulated errors** of KF
- ReID model **does not have many samples**

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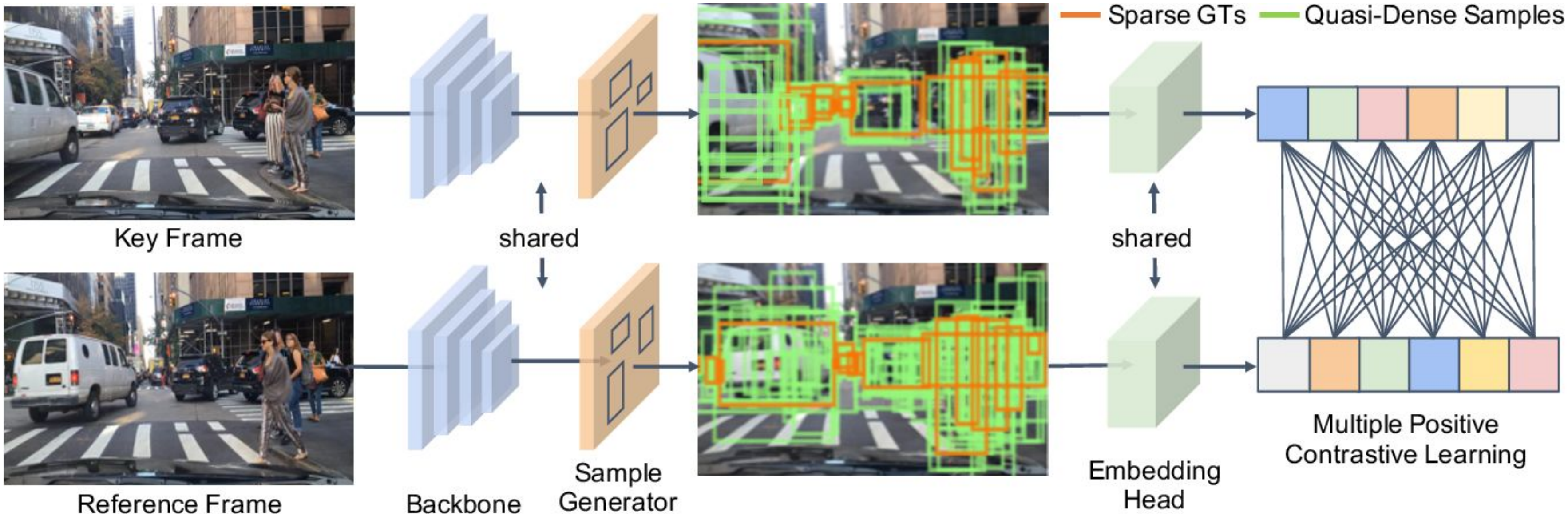
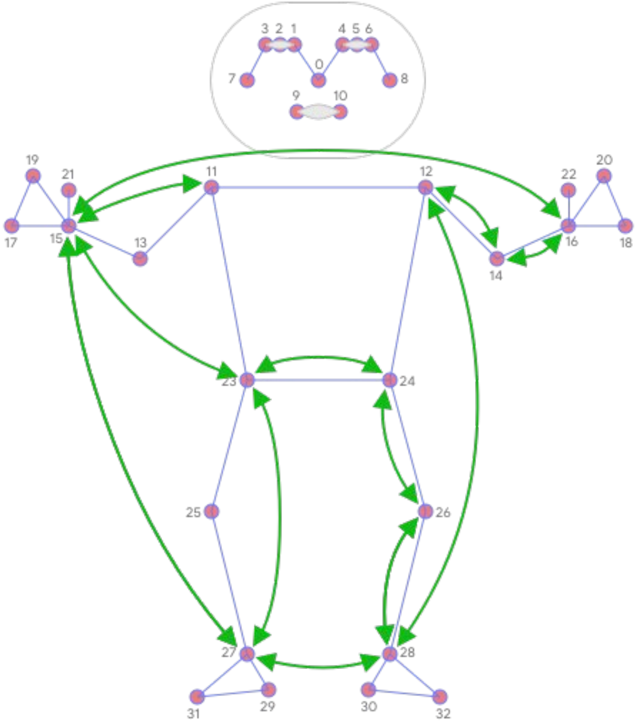
METHODS

3.1 | MOT - QDTrack



Training pipeline of QDTrack

3.2 | MOT - Pose Estimation for ReID



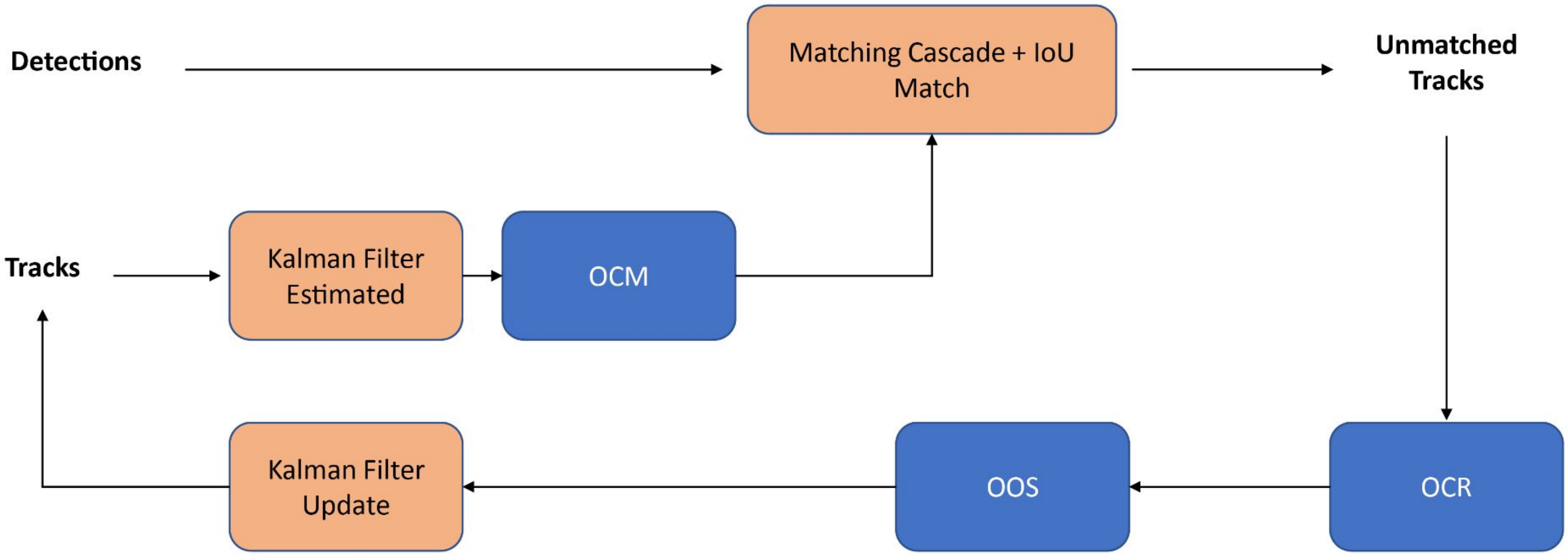
Combine the **features of pose estimation** with the **features of re-identification**:

$$e_{reid-pose} = cat(e_{reid}, e_{pose})$$

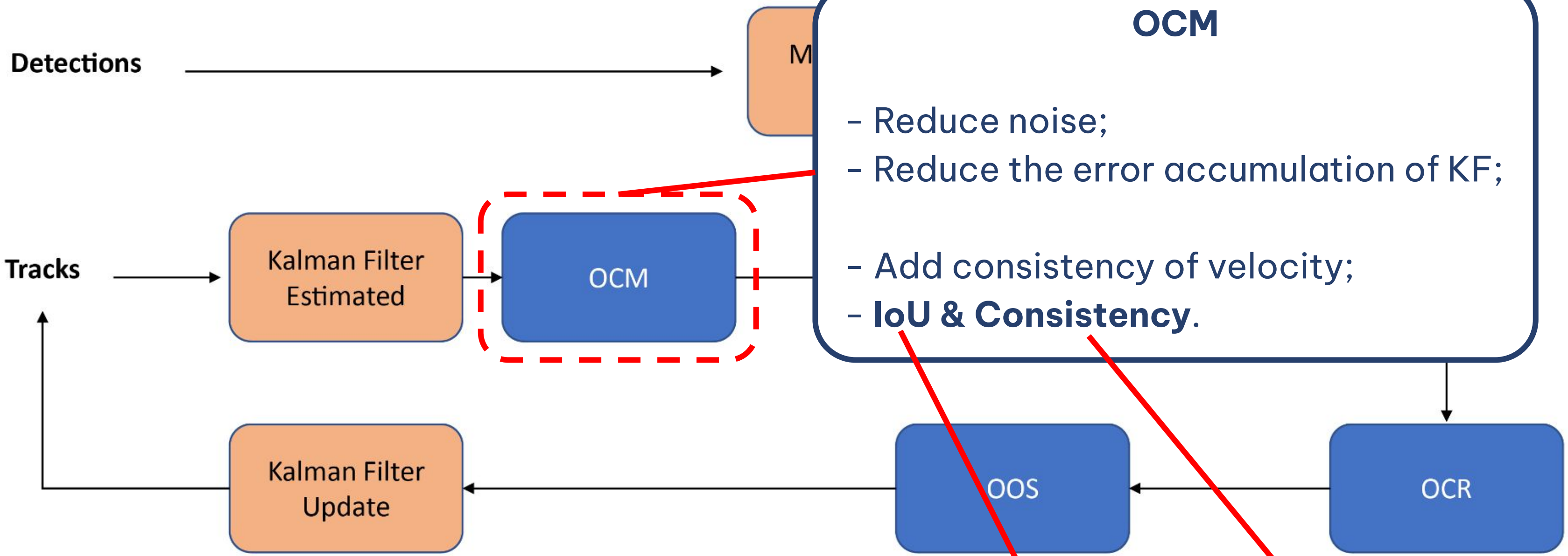
Or

$$d_{i,j} = \alpha \times d_{i,j}^{reid} + (1 - \alpha) \times d_{i,j}^{pose}$$

3.3 | MOT - OCSORT



Adapt OCSORT to DeepSORT pipeline



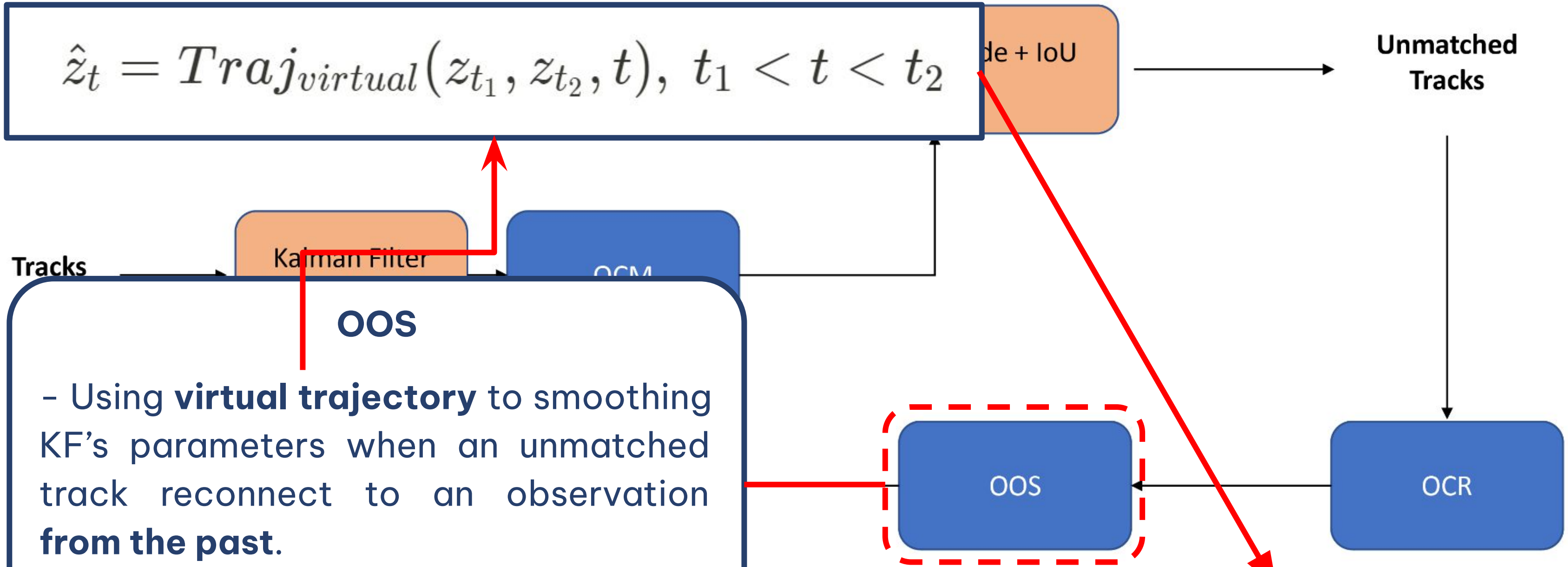
OCM

- Reduce noise;
- Reduce the error accumulation of KF;

- Add consistency of velocity;
- **IoU & Consistency**.

$$C(\hat{X}, Z) = C_{IoU}(\hat{X}, Z) + \lambda C_v(\hat{X}, Z, V)$$

$$\hat{z}_t = Traj_{virtual}(z_{t_1}, z_{t_2}, t),\ t_1 < t < t_2$$



OOS

- Using **virtual trajectory** to smoothing KF's parameters when an unmatched track reconnect to an observation **from the past**.

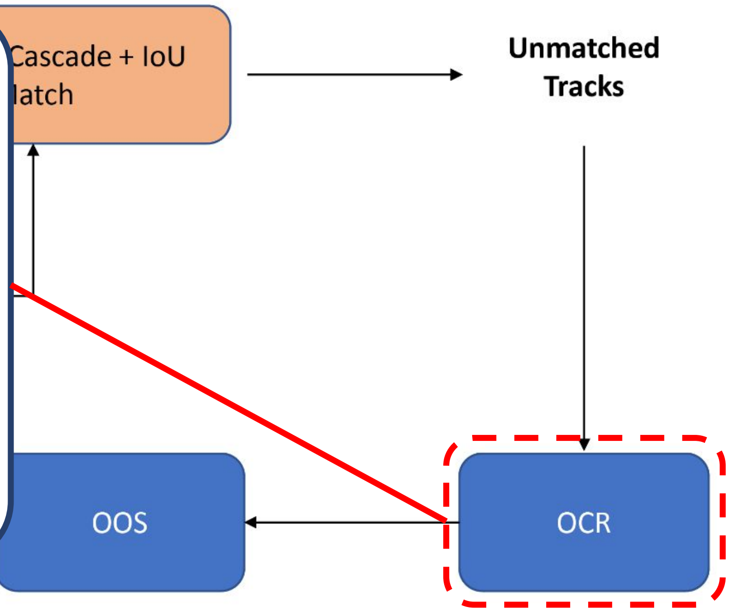
- **Refresh state** estimates as

$$\hat{x}_t = F_t \hat{x}_{t-1} + K_t(\hat{z}_t - H_t F_t \hat{x}_{t-1})$$

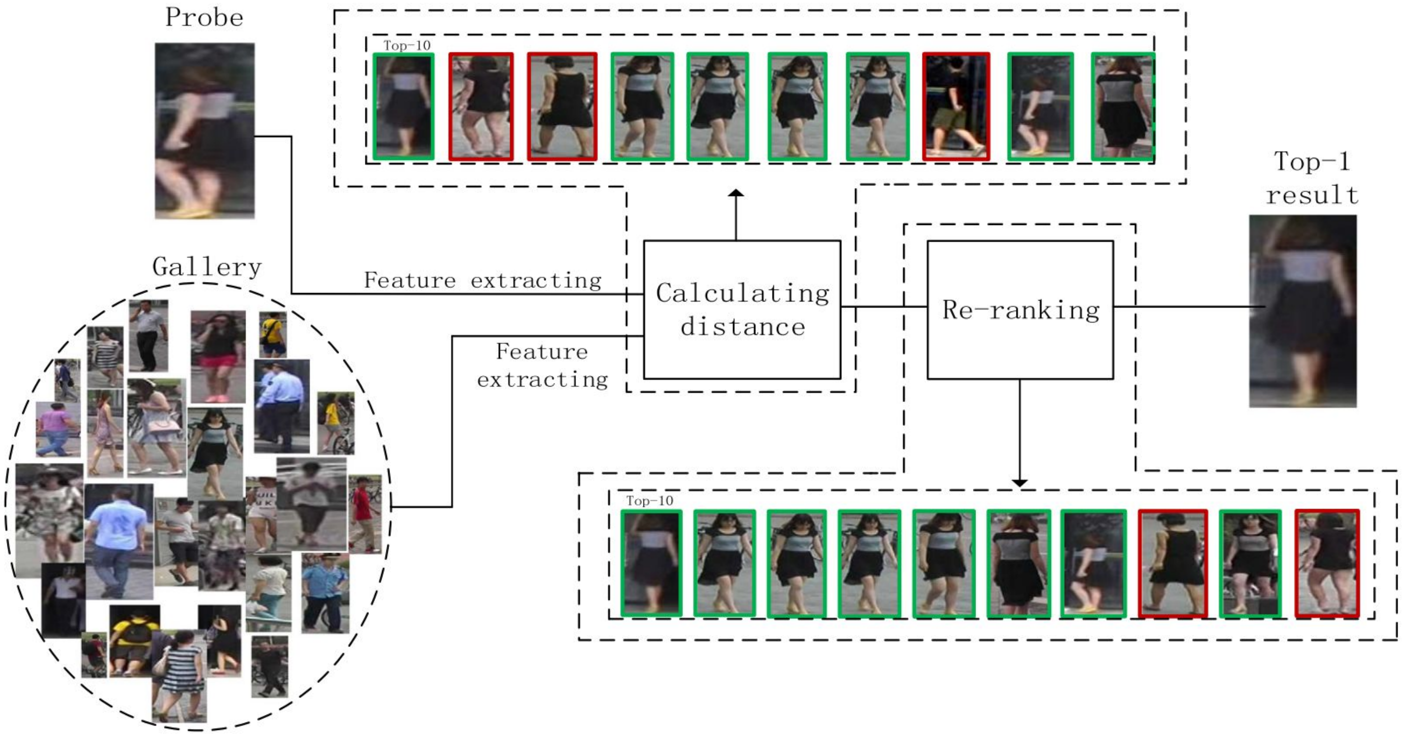
OCR

- Using **IoU Matching** to recovery lost track.
- Matching **last observations** and unmatched tracks.

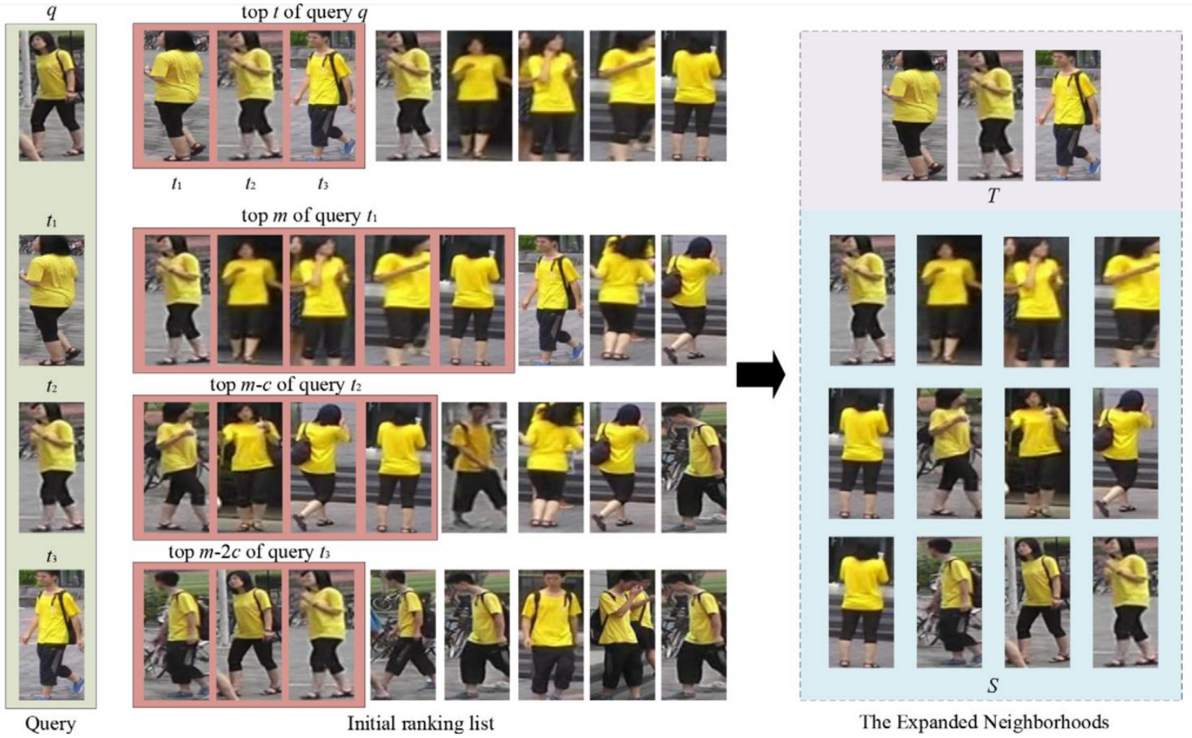
- **After OCR is OOS**, recovered tracks are used to smoothing KF's and reduce error accumulation.



3.4 | MTMC - Re-ranking Algorithm



3.4 | MTMC - Re-ranking Algorithm

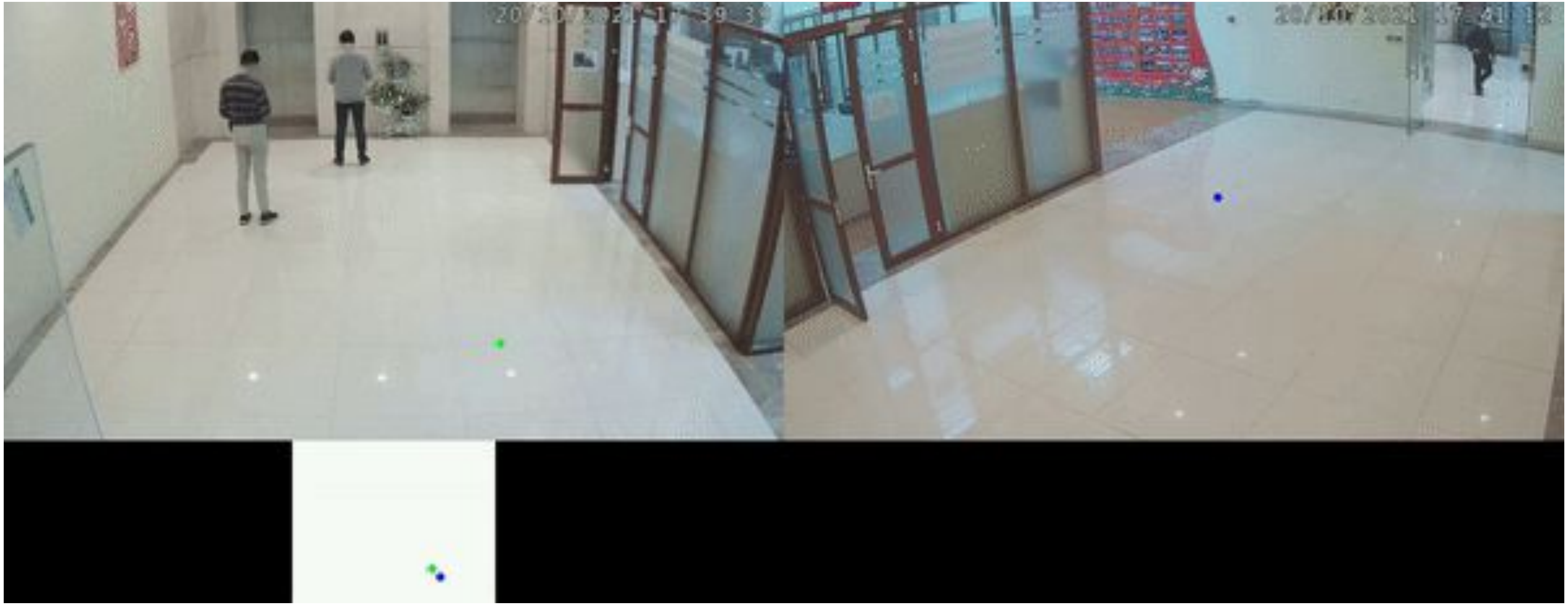


$$M = t + t \times q$$

3.4 | MTMC - Re-ranking Algorithm

$$ECN(p, g_i) = \frac{1}{2M} \sum_{j=1}^{M} d(pN_j, g_i) + d(g_i N_j, p)$$

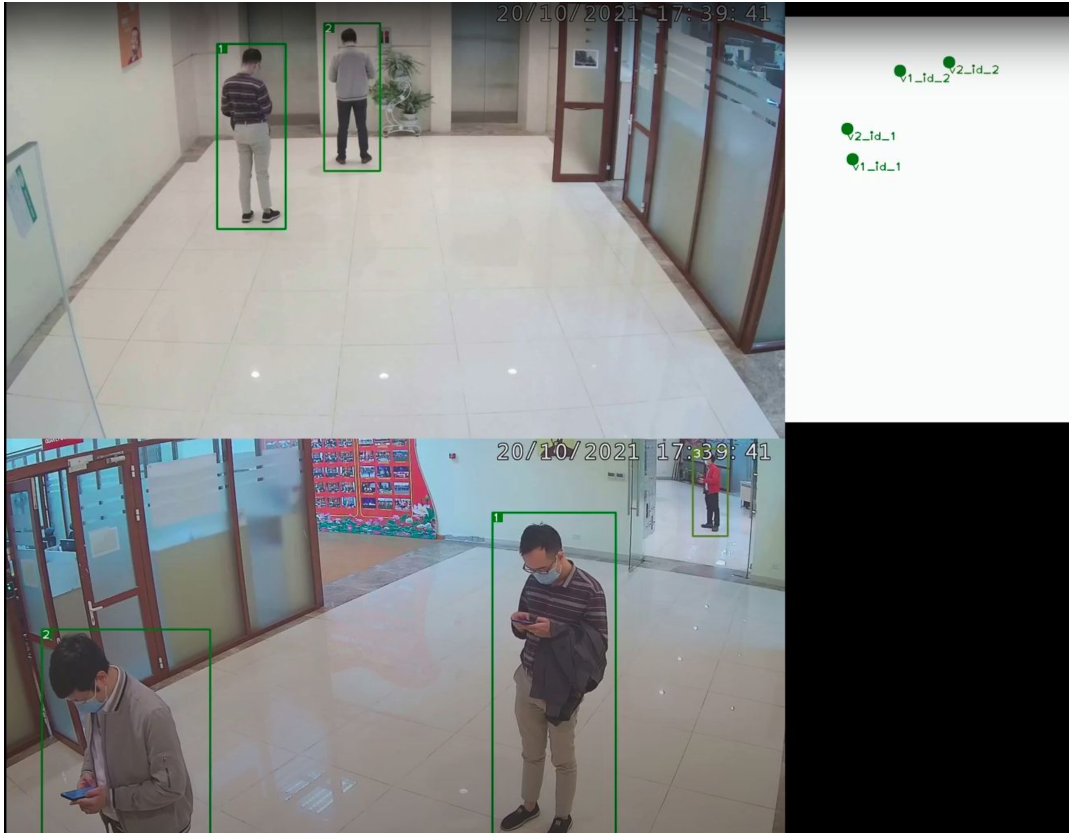
3.5 | MTMC - Spatial Trajectory



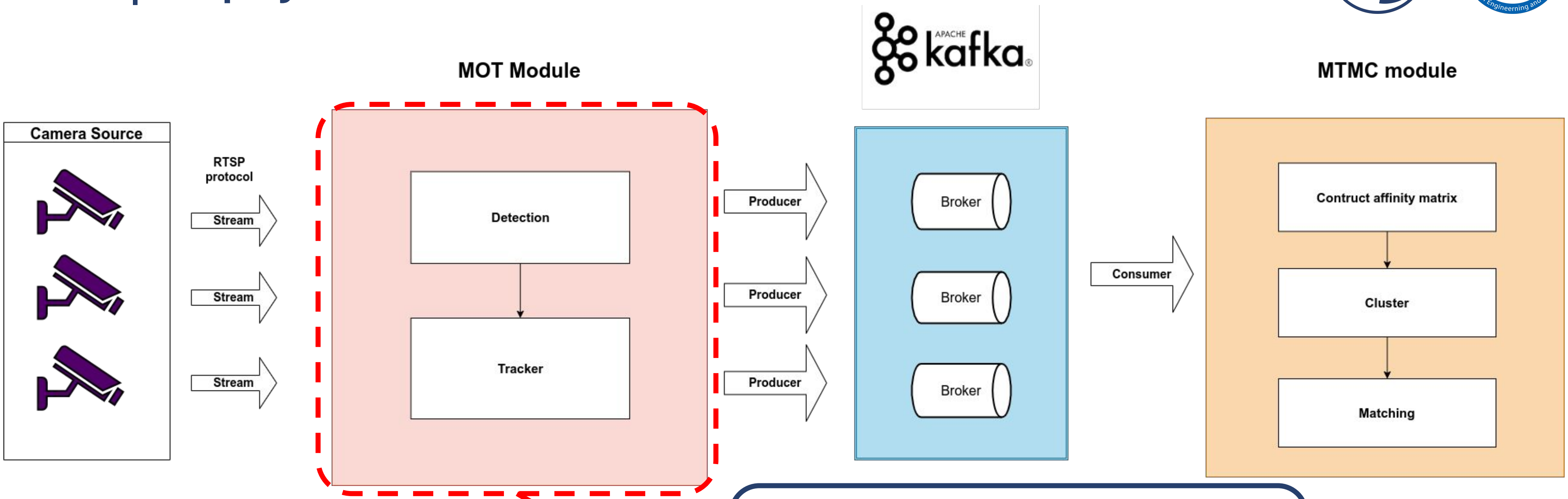
Example of transform camera-view to top-view

3.5 | MTMC - Spatial Trajectory

- Pick **bounding-box center** point to represent object
- Project that point to **top-view map** for every frames
- Using **Haurdoff distance** to calculate distance between set of projected points (trajectory of each object)

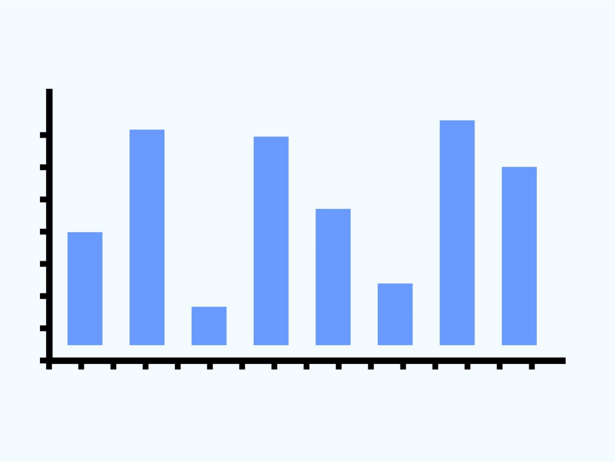


3.6 | Deployments



Deployed using **Deepstream**:

- Source (Decode & Mux)
- Primary inference engine
- Secondary inference engine
- Tee (Split output stream)



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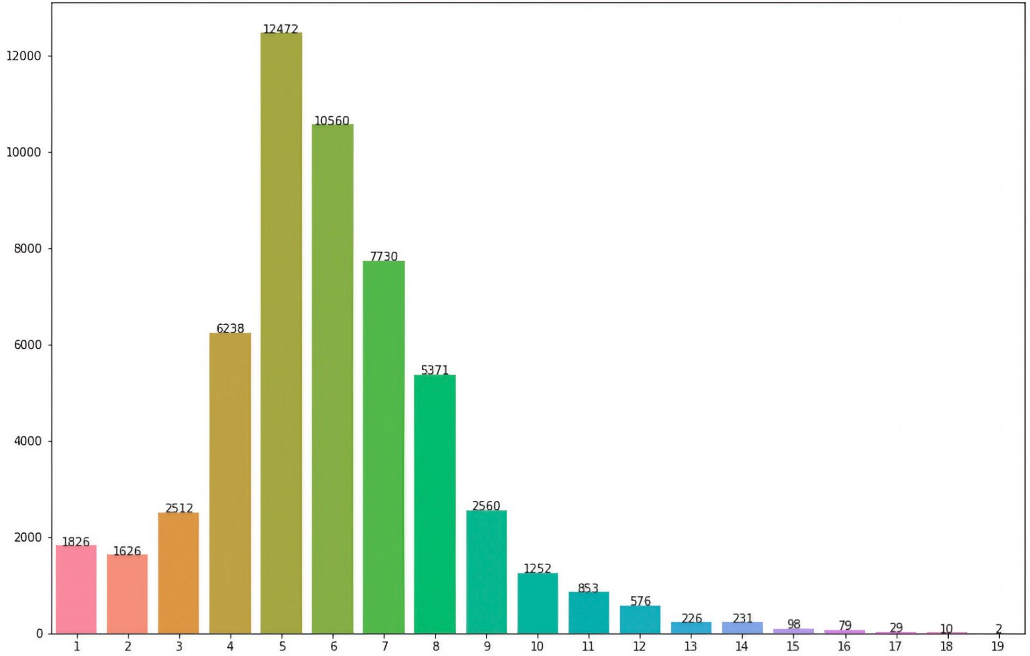
EXPERIMENTS

4.1 | Dataset

- **Overall:**
  - 12 scenes
  - 48 cameras
  - 5 floors
- **Train:**
  - > 530,000 images
  - 1982 unique IDs
- **Evaluation:**
  - ~ 225,500 images
  - 1071 unique IDs



4.2 | Experiments

Model	MOTA	IDF1	IDSW	HOTA
DeepSORT Faster-RCNN	0.638	0.696	3183	57.31
DeepSORT YOLOX	0.705	0.512	9321	47.41
DeepSORT Faster-RCNN w/ Pose Embedding	0.625	0.7	3804	57.36
OCSORT YOLOX	**0.778**	**0.784**	**825**	**67.5**

Result of MOT experiments

Model	MOTA	IDF1
Baseline	0.625	0.443
Baseline + re-ranking	0.709	0.513
Baseline + re-ranking + spatial trajectory	**0.794**	**0.68**

Result of MTMC evaluation on VTX dataset

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CONCLUSIONS & FUTURE WORKS

5.1 | Conclusions

- Combine *Pose estimation* with ReID features ⇒ **Better embedding**
- Increase training samples for ReID using *QDTrack* ⇒ **Better embedding**
- Address *Kalman Filter problem* in state prediction ⇒ **Avoid accumulated errors**
- Adapt *spatial trajectory* ⇒ **Better similarity matrix**
- Achieve **real-time** performance

5.2| Future Works

- Deeper on integrating **human pose information** for MOT phase
- Construct **3D pose** for tracking and matching in 3D view
- Apply **attention** to enhance ReID feature for data association phase
- Improve **camera calibration** process

Thank you for listening!